RAG Learning

3.5 Forces

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs F1–F12

What this topic covers. Identifying and representing the forces acting on a body, resolving forces into components, the conditions for equilibrium, and friction including limiting friction and the angle of friction.

1. The named forces

MEI lists the vocabulary explicitly (spec ref F1).

Force	Description	Direction
Weight	Gravitational pull, $W = mg$	Vertically downwards
Normal reaction	Contact force from a surface	Perpendicular to the surface
Tension	Pull in a string or rod	Along the string, pulling inwards
Thrust / compression	Push in a rod	Along the rod, pushing outwards
Friction	Resists relative sliding	Along the surface, opposing motion or the tendency to move
Resistance	Air resistance, drag	Opposing motion
Driving force	Produced by an engine	Direction of motion

The normal reaction is **not** automatically equal to the weight. Its value depends on the other forces present: pressing down on a block increases it, pulling upwards at an angle reduces it, and on an inclined plane it is $mg\cos\theta$.

A string can only **pull**; a rod can pull (tension) or push (thrust).

2. Force diagrams

Every mechanics problem should begin with a diagram showing **all** the forces on the body — and only those on that body (F3).

An **internal** force acts between parts of a system; an **external** force acts from outside. When a whole system is treated as one body, internal forces cancel in pairs and can be ignored — but they reappear the moment you consider one part on its own.

3. Resolving forces

A force F at angle θ to a chosen direction has components $F\cos\theta$ along it and $F\sin\theta$ perpendicular to it (F6).

Worked example. A force of 20 N acts at 30° above the horizontal.

$$\text{horizontal: } 20\cos 30^\circ = 17.3\text{ N}, \qquad \text{vertical: } 20\sin 30^\circ = 10\text{ N}$$

Choosing directions

On an inclined plane, resolve **parallel and perpendicular to the plane**, not horizontally and vertically. That choice puts the normal reaction along one axis and the friction along the other, so each equation contains fewer unknowns.

For a body on a plane inclined at θ:

$$\text{component of weight down the plane} = mg\sin\theta$$

$$\text{component perpendicular to the plane} = mg\cos\theta$$

Worked example. A 10 kg block rests on a plane inclined at 30°, with $g = 9.8$.
Weight $= 98$ N, so the component down the plane is $98\sin 30^\circ = 49$ N and the normal reaction is $98\cos 30^\circ = 84.9$ N.

$\sin$ goes with the component **along** the slope and $\cos$ with the one **perpendicular** to it. Checking a limiting case settles it: at $\theta = 0$ the plane is flat, the component along it should vanish, and $\sin 0 = 0$ — correct.

4. Equilibrium

A particle is in **equilibrium** if and only if the resultant of the forces acting on it is zero (F5, F7). Equivalently, resolving in *any* direction gives a total of zero.

In practice, resolve in two perpendicular directions to obtain two independent equations, and solve for the unknowns (F9).

Vectors representing forces in equilibrium sum to the zero vector, so drawn head to tail they form a **closed polygon** — a closed triangle for three forces (F8).

Worked example. A crate is pushed at constant velocity across a rough floor. What is the applied horizontal force?
Constant velocity means zero acceleration, so the body is in equilibrium horizontally: the applied force exactly equals the frictional force.

5. Friction

For a body on a rough surface with coefficient of friction μ and normal reaction R:

$$F \leq \mu R$$

Friction takes whatever value is needed to prevent motion, up to a maximum of μR.

When the body is on the point of sliding, the friction is at that maximum — **limiting friction**, $F = \mu R$. The coefficient μ has no units.

Worked example. With $\mu = 0.4$ and $R = 50$ N, the maximum possible frictional force is $0.4 \times 50 = 20$ N. A horizontal push of 15 N produces a frictional force of 15 N and no motion; a push of 25 N exceeds the maximum, so the block slides.

Worked example. A 4 kg block on a rough horizontal surface has $\mu = 0.25$, with $g = 9.8$. Find the force needed to start it sliding.

$$R = 4(9.8) = 39.2 \text{ N}, \qquad F = 0.25 \times 39.2 = 9.8 \text{ N}$$

Use $F = \mu R$ **only** when the body is moving or on the point of moving. For a stationary body not yet at that point, friction equals the applied force, whatever that happens to be — using μR there gives the wrong answer.

The angle of friction

The angle of friction λ satisfies $\tan \lambda = \mu$.

Sliding on a slope

Worked example. Show that a block on a rough plane inclined at θ stays at rest provided $\tan \theta \leq \mu$.
Perpendicular to the plane: $R = mg \cos \theta$.

Parallel to the plane, the component tending to move the block is $mg \sin \theta$. For equilibrium this cannot exceed limiting friction:

$$mg\sin\theta \le \mu R = \mu mg\cos\theta$$

Dividing by $mg\cos\theta$, which is positive for $0 < \theta < 90^\circ$:

$$\tan\theta \le \mu$$

Worked example. A block sits on a rough plane at 20° with $\mu = 0.5$. Does it slide? $\tan 20^\circ \approx 0.364 < 0.5$, so it remains at rest.

Note that the result is independent of the mass — a heavier block is held by proportionally more friction.

Friction on a block in limiting equilibrium on a slope acts **up** the plane, opposing the tendency to slide down. Drawing it downwards is a common diagram error that reverses the whole solution.

Pulling at an angle

Pulling a block with a force angled *above* the horizontal reduces the normal reaction, and therefore reduces the maximum friction — which is why pulling a heavy case is easier than pushing it, since pushing downwards at an angle increases R instead.

6. Gravity

The acceleration due to gravity is not a universal constant but depends on location; on Earth it is modelled as constant, and examinations use $g = 9.8$ unless stated otherwise (F2). The **inverse square law for gravitation is excluded** from this specification.

7. Common pitfalls

- Assuming the normal reaction always equals the weight.
- Swapping $\sin$ and $\cos$ when resolving on an inclined plane.
- Using $F = \mu R$ for a body that is stationary and not on the point of moving.

- Drawing friction in the direction of motion rather than opposing it.
- Omitting a force from the diagram — usually the normal reaction or the weight.
- Resolving horizontally and vertically on a slope, creating unnecessary algebra.
- Treating equilibrium as meaning "at rest" — constant velocity also qualifies.